Practical no. 13

Aim - Data-Driven Testing Using Excel Integration
Write a program to update 10 student records in an Excel file and validate changes on the web application. Perform data reading, writing, and verification using Apache POI library.

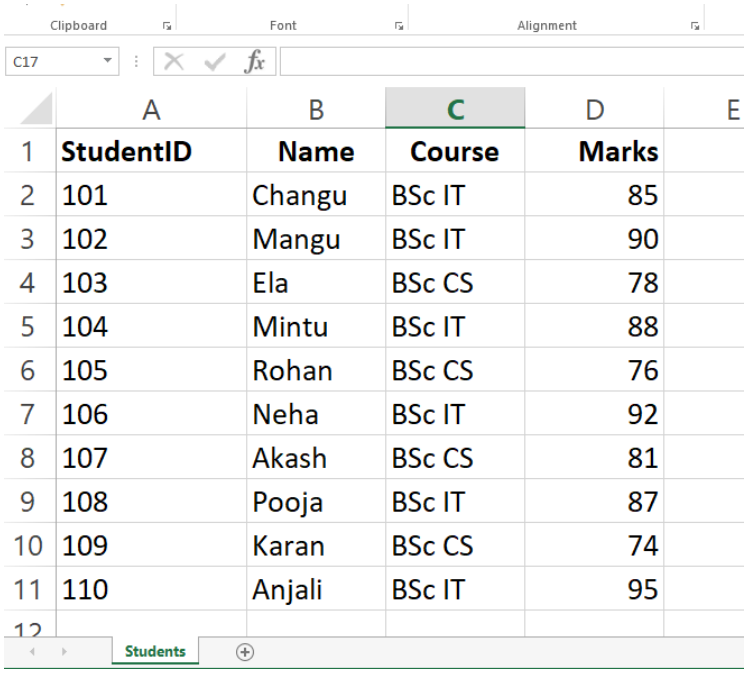
Tool: Selenium WebDriver + Apache POI

Theory –

- **Selenium WebDriver** → Automates the web browser.
  - Opens Chrome.
  - Finds input fields and buttons.
  - Enters student data.
  - Clicks the Update button.
  - Reads and verifies the result.
- **Apache POI** → Reads data from Excel files.
  - Opens StudentData.xlsx.
  - Selects the Students sheet.
  - Reads Student ID, Name, Course, and Marks.
  - Sends that data to Selenium.

Step 1 - Create the project folder
Create: D:\DataDrivenTesting folder.

Step 2 - Create Excel file
Create an Excel workbook.
Create a sheet named: Students
Enter :



	A	B	C	D	E
1	**StudentID**	**Name**	**Course**	**Marks**	
2	101	Changu	BSc IT	85	
3	102	Mangu	BSc IT	90	
4	103	Ela	BSc CS	78	
5	104	Mintu	BSc IT	88	
6	105	Rohan	BSc CS	76	
7	106	Neha	BSc IT	92	
8	107	Akash	BSc CS	81	
9	108	Pooja	BSc IT	87	
10	109	Karan	BSc CS	74	
11	110	Anjali	BSc IT	95	
12					

Students

Save it as:
D:\DataDrivenTesting\StudentData.xlsx

Step 3 - StudentForm.html
Open notepad –

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Management</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 40px;
      background-color: #f4f4f4;
    }
    .container {
      width: 500px;
      background-color: white;
      padding: 25px;
      border: 1px solid #ccc;
      border-radius: 8px;
    }
    h2 {
      color: #333;
    }
    label {
      display: inline-block;
      width: 120px;
      margin-bottom: 15px;
    }
    input {
      width: 250px;
      padding: 8px;
      margin-bottom: 10px;
    }
    button {
      padding: 10px 20px;
      background-color: #007bff;
      color: white;
      border: none;
      cursor: pointer;
      border-radius: 4px;
    }
    button:hover {
      background-color: #0056b3;
    }
    #result {
      margin-top: 20px;
      padding: 15px;
      color: green;
      font-weight: bold;
      border: 1px solid #ccc;
    }
```

```
    </style>
</head>
<body>
    <div class="container">
        <h2>Student Management System</h2>
        <form id="studentForm">
            <label>Student ID:</label>
            <input
                type="text"
                id="studentId">
            <br>
            <label>Student Name:</label>
            <input
                type="text"
                id="studentName">
            <br>
            <label>Course:</label>
            <input
                type="text"
                id="course">
            <br>
            <label>Marks:</label>
            <input
                type="text"
                id="marks">
            <br><br>
            <button
                type="button"
                id="updateButton"
                onclick="updateStudent()">
                Update Student
            </button>
        </form>
        <p id="result"></p>
    </div>
    <script>
        function updateStudent() {
            var id =
                document.getElementById("studentId").value;
            var name =
                document.getElementById("studentName").value;
            var course =
                document.getElementById("course").value;
            var marks =
                document.getElementById("marks").value;
            document.getElementById("result").innerHTML =
                "Student Updated: "
                + id
                + " | "
                + name
```

```
            + " | "
            + course
            + " | "
            + marks;
        }
    </script>
</body>
</html>
```

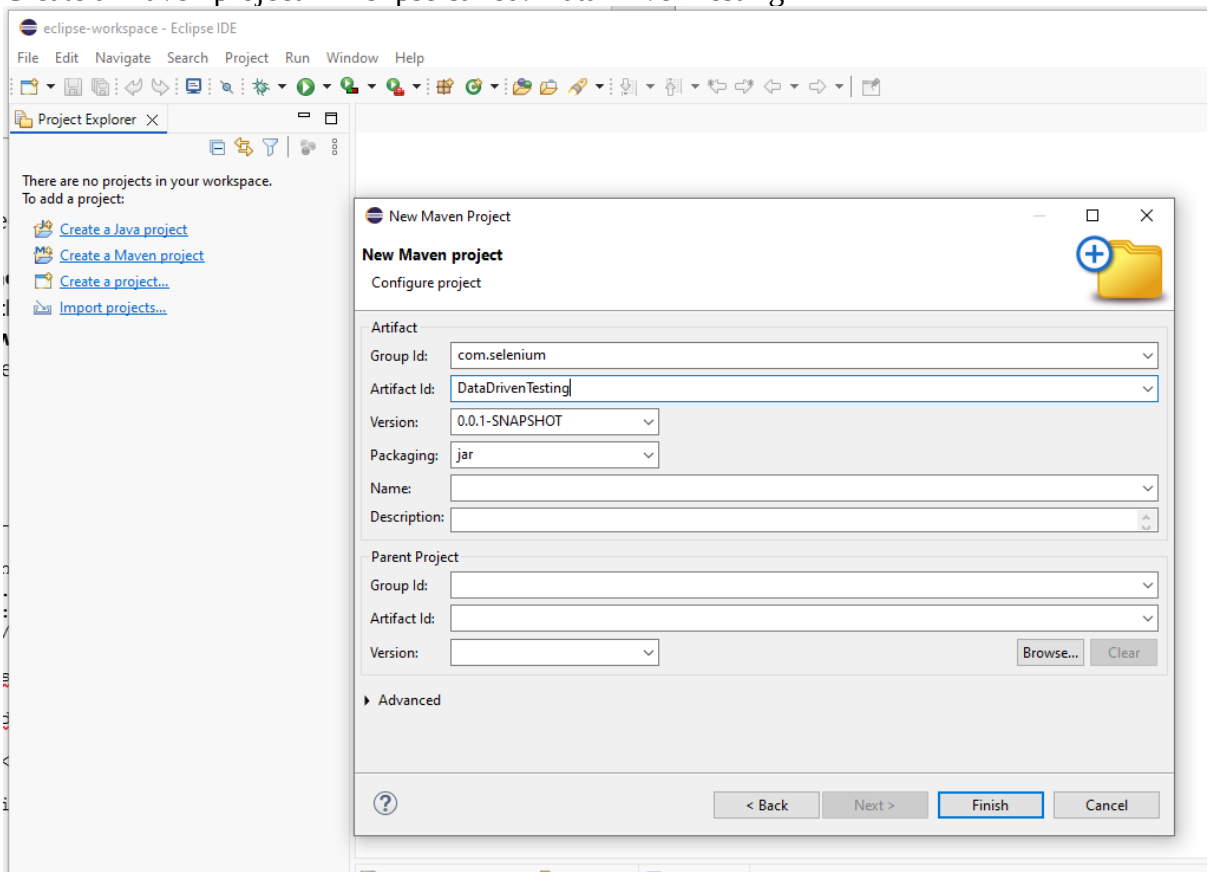
Save - D:\DataDrivenTesting\StudentForm.html

Folder structure

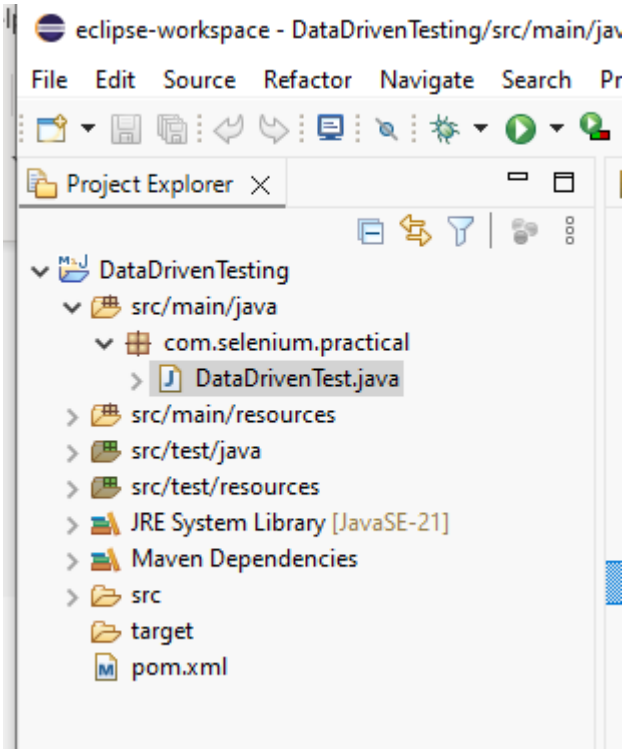
```
D:\DataDrivenTesting
├── StudentData.xlsx
├── StudentForm.html
```

Step 4 - Maven pom.xml

Create a Maven project in Eclipse called: DataDrivenTesting



1. In Eclipse, go to **Window → Show View → Project Explorer**.
2. In Project Explorer, click the **⋮ (three dots)** menu.
3. Look for **Open Project** or **Show All Projects**.
4. If your project is listed but closed, select it and choose **Open Project**.



Use this pom.xml. -

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.selenium</groupId>
    <artifactId>DataDrivenTesting</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <properties>
        <project.build.sourceEncoding>
            UTF-8
        </project.build.sourceEncoding>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
    </properties>
    <dependencies>
        <!-- Selenium WebDriver -->
        <dependency>
            <groupId>
                org.seleniumhq.selenium
            </groupId>
            <artifactId>
                selenium-java
            </artifactId>
            <version>4.35.0</version>
        </dependency>
        <!-- Apache POI -->
```

```
        <dependency>
            <groupId>
                org.apache.poi
            </groupId>
            <artifactId>
                poi-ooxml
            </artifactId>
            <version>5.4.1</version>
        </dependency>
    </dependencies>
</project>
```

After saving:

Right-click project → Maven → Update Project → OK

Step 5 - DataDrivenTest.java

Go to: src/main/java Right-click: New → Package Create: com.selenium.practical

Then right-click the package: New → Class Class name: DataDrivenTest

Create:

```
src/main/java
    └── com.selenium.practical
            └── DataDrivenTest.java
```

Use this complete program:

DataDrivenTest.java -

```
package com.selenium.practical;
import java.io.File;
import java.io.FileInputStream;
import org.apache.poi.ss.usermodel.DataFormatter;
import org.apache.poi.ss.usermodel.Row;
import org.apache.poi.ss.usermodel.Sheet;
import org.apache.poi.ss.usermodel.Workbook;
import org.apache.poi.ss.usermodel.WorkbookFactory;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class DataDrivenTest {
    public static void main(String[] args) {
        WebDriver driver = null;
        FileInputStream file = null;
        Workbook workbook = null;
        try {
            // Excel file path
            String excelPath = "D:\\DataDrivenTesting\\StudentData.xlsx";
            // HTML file path
            String htmlPath = "file:///D:/DataDrivenTesting/StudentForm.html";
            // Check Excel file
            File excelFile = new File(excelPath);
```

```
        if (!excelFile.exists()) {
            throw new RuntimeException("Excel file not found: " + excelPath);
        }
        // Open Excel file
        file = new FileInputStream(excelFile);
        System.out.println("Excel file opened successfully.");
        // Create workbook
        workbook = WorkbookFactory.create(file);
        // Select Students sheet
        Sheet sheet = workbook.getSheet("Students");
        if (sheet == null) {
            throw new RuntimeException("Students sheet not found.");
        }
        System.out.println("Students sheet selected successfully.");
        // Create data formatter
        DataFormatter formatter = new DataFormatter();
        // Launch Chrome
        driver = new ChromeDriver();
        driver.manage().window().maximize();
        System.out.println("Chrome launched successfully.");
        // Set implicit wait
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
        // Create explicit wait
        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
        // Open student application
        driver.get(htmlPath);
        System.out.println("Student Management System opened.");
        // Process 10 student records
        for (int i = 1; i <= 10; i++) {
            // Get Excel row
            Row row = sheet.getRow(i);
            if (row == null) {
                System.out.println("Row " + i + " is empty. Skipping.");
                continue;
            }
            // Read student ID
            String studentId = formatter.formatCellValue(row.getCell(0));
            // Read student name
            String name = formatter.formatCellValue(row.getCell(1));
            // Read course
            String course = formatter.formatCellValue(row.getCell(2));
            // Read marks
            String marks = formatter.formatCellValue(row.getCell(3));
            System.out.println();
            System.out.println("Processing Student: " + studentId);
            // Enter student ID
            WebElement studentIdField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("studentId")));
            studentIdField.clear();
            studentIdField.sendKeys(studentId);
```

```
            // Enter student name
            WebElement studentNameField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("studentName")));
            studentNameField.clear();
            studentNameField.sendKeys(name);
            // Enter course
            WebElement courseField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("course")));
            courseField.clear();
            courseField.sendKeys(course);
            // Enter marks
            WebElement marksField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("marks")));
            marksField.clear();
            marksField.sendKeys(marks);
            // Click Update Student button
            WebElement updateButton =
wait.until(ExpectedConditions.elementToBeClickable(By.id("updateButton")));
            updateButton.click();
            // Read result
            WebElement resultElement =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("result")));
            String result = resultElement.getText();
            System.out.println("Web Result: " + result);
            // Verify student data
            if (result.contains(studentId) && result.contains(name) && result.contains(course)
&& result.contains(marks)) {
                System.out.println("PASS - Student " + studentId + " data verified
successfully.");
            } else {
                System.out.println("FAIL - Student " + studentId + " data verification failed.");
            }
            // Pause before next record
            Thread.sleep(500);
        }
        // All records processed
        System.out.println();
        System.out.println("All 10 student records processed.");
    } catch (Exception e) {
        // Handle exception
        System.out.println();
        System.out.println("An error occurred:");
        e.printStackTrace();
    } finally {
        // Close browser
        if (driver != null) {
            driver.quit();
            System.out.println("Browser closed successfully.");
        }
        // Close workbook
```

```
            if (workbook != null) {
                try {
                    workbook.close();
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
            // Close Excel file
            if (file != null) {
                try {
                    file.close();
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        }
    }
}
```

Step 6 - Run

In Eclipse:

Right-click DataDrivenTest.java

↓

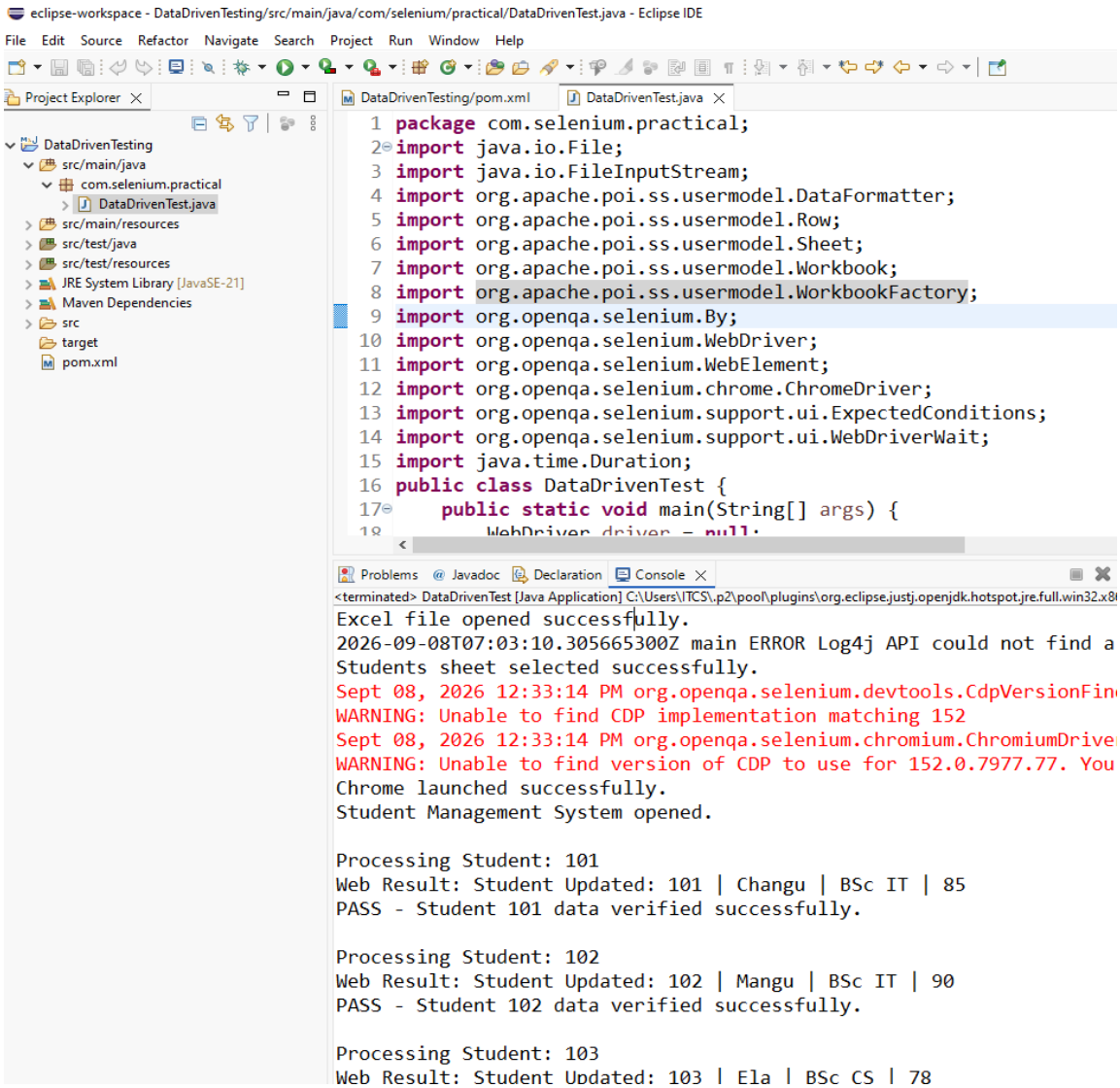
Run As

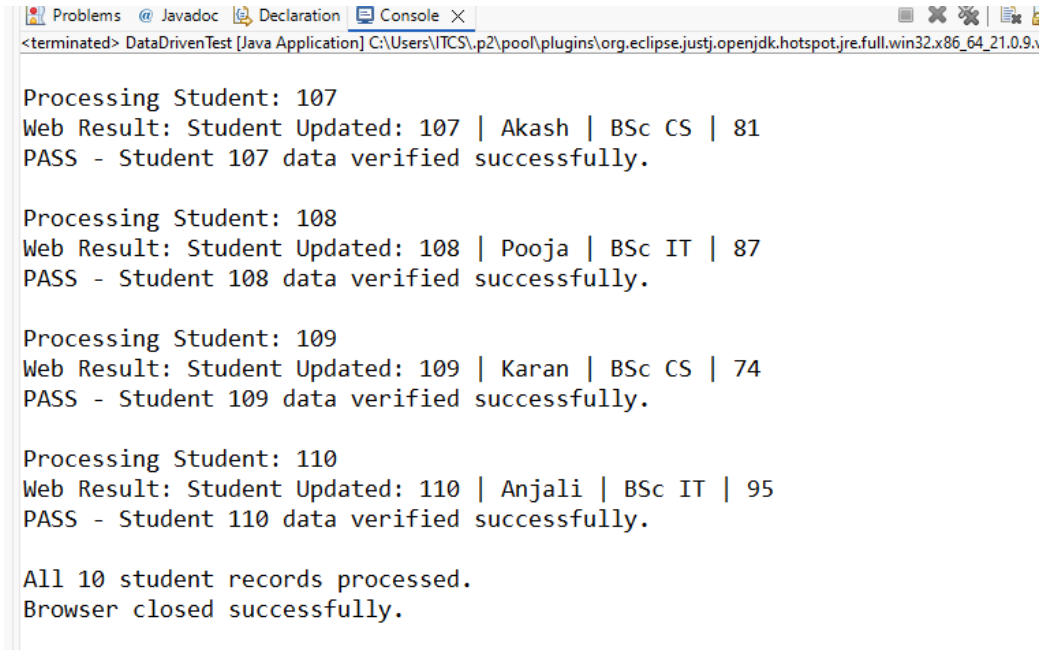
↓

Java Application

Chrome will automatically process all 10 records.

Expected output:


eclipse-workspace - DataDrivenTesting/src/main/java/com/selenium/practical/DataDrivenTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Project Explorer
DataDrivenTesting
src/main/java
com.selenium.practical
DataDrivenTest.java
src/main/resources
src/test/java
src/test/resources
JRE System Library [JavaSE-21]
Maven Dependencies
src
target
pom.xml
DataDrivenTesting/pom.xml
DataDrivenTest.java
1 package com.selenium.practical;
2 import java.io.File;
3 import java.io.FileInputStream;
4 import org.apache.poi.ss.usermodel.DataFormatter;
5 import org.apache.poi.ss.usermodel.Row;
6 import org.apache.poi.ss.usermodel.Sheet;
7 import org.apache.poi.ss.usermodel.Workbook;
8 import org.apache.poi.ss.usermodel.WorkbookFactory;
9 import org.openqa.selenium.By;
10 import org.openqa.selenium.WebDriver;
11 import org.openqa.selenium.WebElement;
12 import org.openqa.selenium.chrome.ChromeDriver;
13 import org.openqa.selenium.support.ui.ExpectedConditions;
14 import org.openqa.selenium.support.ui.WebDriverWait;
15 import java.time.Duration;
16 public class DataDrivenTest {
17 public static void main(String[] args) {
Problems Javadoc Declaration Console
<terminated> DataDrivenTest [Java Application] C:\Users\ITCS\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x8
Excel file opened successfully.
2026-09-08T07:03:10.305665300Z main ERROR Log4j API could not find a
Students sheet selected successfully.
Sept 08, 2026 12:33:14 PM org.openqa.selenium.devtools.CdpVersionFin
WARNING: Unable to find CDP implementation matching 152
Sept 08, 2026 12:33:14 PM org.openqa.selenium.chromium.ChromiumDrive
WARNING: Unable to find version of CDP to use for 152.0.7977.77. You
Chrome launched successfully.
Student Management System opened.
Processing Student: 101
Web Result: Student Updated: 101 | Changu | BSc IT | 85
PASS - Student 101 data verified successfully.
Processing Student: 102
Web Result: Student Updated: 102 | Mangu | BSc IT | 90
PASS - Student 102 data verified successfully.
Processing Student: 103
Web Result: Student Updated: 103 | Ela | BSc CS | 78


Problems Javadoc Declaration Console
<terminated> DataDrivenTest [Java Application] C:\Users\ITCS\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.
Processing Student: 107
Web Result: Student Updated: 107 | Akash | BSc CS | 81
PASS - Student 107 data verified successfully.
Processing Student: 108
Web Result: Student Updated: 108 | Pooja | BSc IT | 87
PASS - Student 108 data verified successfully.
Processing Student: 109
Web Result: Student Updated: 109 | Karan | BSc CS | 74
PASS - Student 109 data verified successfully.
Processing Student: 110
Web Result: Student Updated: 110 | Anjali | BSc IT | 95
PASS - Student 110 data verified successfully.
All 10 student records processed.
Browser closed successfully.